Towards a new paradigm for brain-like sequential computations
[working document]

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Abstract

A global technological and scientific race is underway with the goal of transcending traditional digital computing principles and replicating the brain's adaptable multi-scale spatio-temporal analog computations. Driving this race are the limitations of digital computing principles (non-robustness to noise, speed limits, energy greed) and the technological substrates (integrated circuits) in which these principles are laid on, approaching their physical limits of miniaturisation. Through millions of years of evolution, the brain has solved these problems, yet its computational principles have still not been deciphered. To unravel it, we propose a new paradigm based on *multi-scale excitable far-from-equilibrium systems with dissipative structures*, where timescale competition between variables create regions of phase space called *funnels*, geometrically akin to bow-tie funnels or bottlenecks with both contracting and expanding directions. Funnels can be seen as novel computational elements that control trajectories into desired phase-space directions, thus enabling the design of computations that are robust to noise yet sensitive to stimuli, and memory representation. Significantly, these new timescale competition principles can be related to competing physical processes in analog circuits, unlocking the gate to analog neuromorphic implementations.

1 Introduction

The brain is arguably the most formidable complex far-from-equilibrium system that orchestrates information across multiple spatio-temporal scales and mediates cognition. It poses daunting scientific challenges, with numerous open questions representing the frontiers of our understanding. The ramifications in solving these challenges are beyond immediate reach, but in the long term will have unprecedented social, medical, technological, scientific and economical impacts. Specifically, the present article addresses an enigmatic open problem at the interface between Neuroscience, Computer Science and Neuromorphic Computing, which can be formulated as follows: *"How does the brain realize efficient Turing-like analog computations that are paradoxically simultaneously robust to noise and sensitive to stimuli, hence effectively beyond Turing?"* As a corollary: *"How does the brain construct highly functional neuronal networks capable of learning (e.g. symbolic structures and computations) and of memory representation, with imperfect neurons and synapses?"* A number of key insights and scientific advancements, combined with recent results in multiscale excitable far-from-equilibrium systems [Desroches et al., 2012, Desroches et al., 2013, Desroches et al., 2016, Beim Graben et al., 2019], enable us to propose and discuss here a pathway towards answering this question.

First of all, recent advances in Neuroscience show that the brain realises sequential Turing-like computations via metastable states [Beim Graben et al., 2019], and that these computations are both robust to noise and sensitive to stimuli [Mazor and Laurent, 2005, Jones et al., 2007]. Even though the brain functions in a noisy environment with extreme trial-to-trial variability of neuronal responses, an embedding in phase space, as well as hidden Markov model extraction, do reveal robust sequential neuronal activations; see, e.g., [Mazor and Laurent, 2005, Fig. 4] for examples of such sequential dynamics in sensory cortical data.

Within the realm of digital computers, computations are not robust to noise. Indeed, *Turing machines (TMs)* [Turing, 1936] where inspired by how the Brain computes: a TM executes the exact set of sequential computations, given an initial condition (i.e. fixed input data on a tape), and any small perturbation, either on the initial data or set of instructions, results in a corrupted computation. This led Turing to search further as he likely struggled with the concept that the brain (as a biological electro-chemical process) is capable of creating robust far-from-equilibrium complex spatial-temporal structures that are in fact computing machines (i.e., algorithms). These ideas are exposed in his two seminal papers: (i) *"Intelligent Machinery"* (1948) [Turing, 1948] describes neuron-like computing and shows that he anticipated Artificial Intelligence (AI) and Connectionism; (ii) *"The chemical basis of morphogenesis"* (1952) [Turing, 1952].

Unprecedented breakthroughs in AI have vastly affected this area of research. Ever since Hebb's work on learning and memory [Hebb, 1949] and Rosenblatt's *perceptron* [Rosenblatt, 1957] leading to modern-day AI systems such as deep neural networks [Bengio et al., 2009, LeCun et al., 2015], echo-state networks [Jaeger, 2001] or liquid-state machines [Maass et al., 2002], these breakthroughs have raised a dichotomy: *data-centric computation* vs *program-centric computation*. In data-centric computations, the machine learns by sampling from a high-dimensional set of finite data (e.g., handwriting) and subsequently makes predictions (e.g., handwriting recognition). This is seemingly impossible to achieve in *program-centric computations*, that is, via an algorithm rendered by a programmer. However, it is conceivable that Turing-type computations (e.g., *Universal Turing Machines/UTMs*, which process algorithms) may learn recurrent structures and symbolic

computations due to their recursive mathematical properties. In contrast, *deep neural networks* have limited capacity in learning symbolic computations, even though the recent surge of generative AI engine such as *chatGPT* are pushing the limit of machine learning [Noy and Zhang, 2023]. Moreover, it is an open problem to explain exactly how deep neural networks make predictions and this has led to the new sub-field of *explainable AI* [Fellous et al., 2019]. Thus, it is a major challenge to exhibit a system capable of learning both symbolic structures and computations as well as explaining the learnt patterns, although some steps (albeit very limited) were achieved by *Google's Turing Neural Networks* [Graves et al., 2014]. An alternative approach is our recent work on *Neural Automata (NA)*, which map, via *algebraic representation theory*, TMs/UTMs to dynamical systems/recurrent neural networks. Our proof-of-concept shows that NA are capable of learning symbolic computations and are explainable [Carmantini et al., 2017] and we recently extended this approach by computing invariants for NA [Uria-Albizuri et al., 2023].

In the field of Computational Neuroscience, there is a dichotomy between brain-like and digital computations. On the one hand, both the 1963 Nobel Prize winning neuronal spiking model of Hodgkin and Huxley [Hodgkin and Huxley, 1952] and 2024 Nobel Prize winning attractor neural networks for memory representation by Hopfield [Hopfield, 1982], support the hypothesis that *the brain is a complex analog dynamical system*. On the other hand, IBM's cat-scale cortical simulation of 1 billion neurons and 10 trillion synapses [Ananthanarayanan et al., 2009] is not amenable to mathematical explanation and it highlights the inefficiency of von Neumann computer architectures in terms of energy consumption and speed limits due to *von Neumann bottleneck* [Chen et al., 2015]. Namely, instruction fetch and data operations cannot occur simultaneously due to shared bus. Indeed, running at 1/83 real-time, it required 144 TB of memory and 147,456 CPUs operating at 20 Watts/CPU (i.e., 2.9 MW).

Limitations of digital computations have boosted Neuromorphic Computing. Indeed, this has led to the search of novel materials and architectures (e.g., *in-memory computation* [Chen et al., 2015]) for brain-like computations. Initial attempts on neuromorphic systems were based on *Memistors* (1960) [Widrow, 1960], an electrochemical plating of carbon rods that emulate synapses. Significantly, neuromorphic computing was set in firm ground when in 1971 Chua postulated on the basis of symmetry arguments the existence of a missing fourth circuit element, which he called *Memristor* (*memory resistor*: the resistance depends on the integral of the input applied to the terminals) [Chua, 1971]. Since the conductivity of memristive materials is adaptable by electrical stimulation, memristors can potentially mimic neurons and synapses. Indeed, several memristive devices have been proposed [Yang et al., 2008, Stewart et al., 2004, Kozicki et al., 2006, Szot et al., 2006, Dong et al., 2007, Tsubouchi et al., 2007, Merolla et al., 2014]. Neuromorphic computing is a very active research area with many current directions under consideration [Christensen et al., 2022], and where the concept of multi-timescale dynamics is slowly making a dent [Jaeger and Catthoor, 2023, Jaeger et al., 2023]. This is in line with the pathway we are sketching in the present perspective article.

Mathematical Neuroscience offers an hypothesis to explain brain computations. Mathematicians have sought biologically plausible mathematical objects to enable the design of complex yet mathematically explainable neuronal circuits that realize analog computations and memory representation. The current leading hypothesis and framework for brain computation is the so-called *heteroclinic sequences* (a.k.a. *winnerless competition*) [Rabinovich et al., 2008, Tsuda, 1991]. Heteroclinic sequences refer to a connected network of saddle-type attractors (i.e., metastable states) that emerge within the phase space of a dynamical system. They can explain Turing-like computations since systems's trajectories (close to *heteroclinic orbits*) flow sequentially between adjacent attractors (under external inputs) by leaving the vicinity of an attractor along its unstable manifold and approaching another attractor following its stable manifold. However, surprisingly textitheteroclinic sequences have never been tested for *real-world neuromorphic implementations*. Thus, we suggest to the community to validate (with engineers) the following features: (i) *robustness to noise*; (ii) *scalability to large networks*; (iii) *do the symmetries, invariants subspaces of heteroclinic networks impose limitations on heterogeneity?* (e.g., parameters and node equations). This is relevant since fabrication processes impose no identical circuit elements; (iv) A heteroclinic orbit that approaches a saddle attractor via its stable manifold takes an *infinite amount of time* (in mathematical terms) to reach the attractor. In real-world circuits infinite processes are not feasible. However, in a real-world neuromorphic implementation, this could be due to a saddle-type slow manifold instead a real saddle induced by fabrication imperfections and drifts of various latent variables. In any case, this remains to be tested on real circuits. Note that a variant to heteroclinic sequences is provided by the concept of chaotic itinerancy [Inoue et al., 2020], but it yields the same questions regarding practical implementation in analog circuits.

More recent approaches include *excitable networks* and their applications [Ashwin and Postlethwaite, 2016, Ashwin and Postlethwaite, 2021, Ashwin et al., 2024, Postlethwaite et al., 2024], however yet without any practical analog implementation either. Nevertheless, such networks show promising enough results on computers that they would deserve to be tested in an analog environment. Finally, a recent preprint [Cotler and Rezchikov, 2024] reviews computations achieved with dynamical systems, which further motivates the work programme behind this short note.

2 Proposed new paradigm for robust brain computations

Our research has recently attempted to overcome the limitations of the heteroclinic sequences paradigm. These first attempts preserve some of the heteroclinic structure but introduce a different type of dynamics that trigger the switch between successive saddle attractors, namely via the use of *dynamic bifurcations* [Desroches et al., 2012, Desroches, 2015]; they are henceforth denoted as hybrid *heteroclinic-dynamic sequence/HDS*. This approach does not require symmetry, it is less prone to numerical instabilities, however its dependence on attractors casts doubts on its real-world implementability. This has led us to reconsider his earlier work on a well-established mathematical object termed *multi-scale funnel*, which he was the first to compute [Desroches et al., 2008a]. Indeed, we recently realized that it can be used to design biologically plausible neuromorphic circuits capable of brain-like computations and memory representations. Hence he proposes a new computing

principle based on multi-scale excitable and far-from-equilibrium systems with dissipative structures (in the sense of 1977 Chemistry Nobel laureate Ilya Prigogine), where competition of timescales between system variables gives emergence to (controllable) regions of phase space that correspond to *funnels* (geometrically akin to bow-tie funnels) with both contracting and expanding directions [Desroches et al., 2012, Desroches et al., 2008a].

Funnels can be seen as novel computational elements that control system trajectories by first attracting them and subsequently ejecting them into desired directions of phase space. In this respect, funnels mimic saddle attractors of heteroclinic sequences, yet they are not attractors. Funnels enable computations that are both robust to noise and sensitive to stimuli, since perturbations of trajectories passing through a given funnel will still correspond to the same computation. They account for adaptive neurophysiological processes, such as plasticity, as recently shown by the authors [Rodrigues et al., 2016], non-synaptic plasticity [Gidon et al., 2020, Mozzachiodi and Byrne, 2010] and excitability. Crucially, it enables the design of biologically plausible analog circuits. To prove this, we have recently developed a real analog circuit prototype (with standard electronics) exhibiting a single funnel (unpublished); see Fig. 1(B). Moreover, funnels can emerge in *piecewise-linear/PWL slow-fast models*, as shown by the authors in [Desroches et al., 2016] and in this context the control of funnels is further simplified, mediated by only two parameters that regulate the ratio contraction/expansion.

This implies that funnels can be implemented with memristive devices, since memristive devices are modelled via PWL models [Parihar et al., 2018]. Significantly, this opens the pathway towards scalable and energy efficient funnel-based neuromorphic networks. To advance this new direction, our future objective is to consolidate and lay the foundations of the proposed paradigm via the following overarching aim: we plan to demonstrate a new paradigm for brain-like analog computations based on funnels, which are novel computational elements that emerge in multi-scale excitable far-from-equilibrium systems. This novel concept will be deployed onto next-generation analog neuromorphic circuits. In doing so, we wish to achieve computations that are robust to noise yet sensitive to stimuli, as well as capable of memory representation (e.g. associative memory).

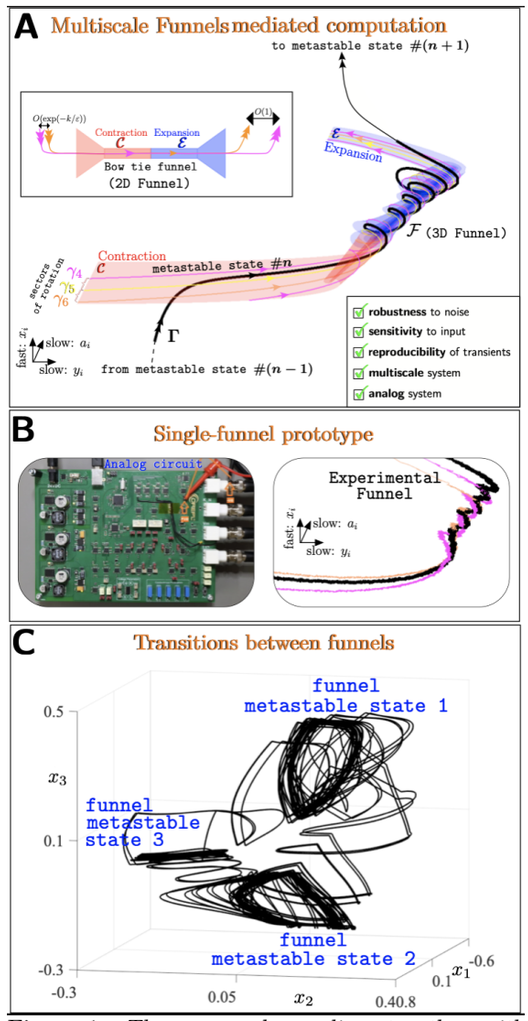


Figure 1: *The proposed paradigm together with preliminary results.*

3 Discussion

A natural perspective would be to investigate theoretically and computationally how to embed networks of funnel systems into neural-network architecture. The approach should be constructive and, hence, explainable. The idea of the present article is to stimulate mathematical neuroscientists to develop a full-fledged alternative theory for brain-like computations beyond what already exists in the literature, such as heteroclinic sequences and excitable networks. Importantly, such an alternative theory, for which we advocate the use of multi-timescale dynamics and funnels, should pass the test of implementability in real experiments. Noteworthy, funnel-based computations may well not be the final answer, we are simply proposing this paradigm in order to stimulate further discussions and the development of novel theories.

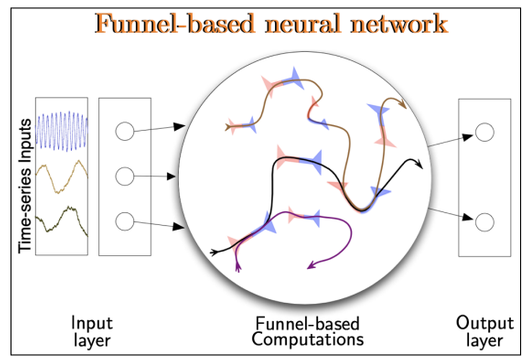


Figure 2: *Funnel-based network embedded within a putative neural-network architecture.*

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A Model systems for funnel dynamics

A.1 Simple funnel system in the self-coupled FitzHugh-Nagumo model

A prototypical neuro-inspired funnel system is standard FitzHugh-Nagumo (FHN) model augmented by a slow dynamic self-coupling, that is, a slow autapse [Desroches et al., 2008b]. It represents an idealised neuron model with a self-connection, that is, an autapse. Its equations are given by

$$\begin{aligned} \dot{v} &= v - v^3/3 - w + \gamma s v \\ \dot{w} &= \varepsilon(av + bw) \\ \dot{s} &= \varepsilon H(v)(\beta v + \delta w) - s \end{aligned} \tag{1}$$

A.2 Three-funnel system

We consider a micro-circuit formed by three inhibitory-coupled FHN models with inhibitory connections, as defined in [Rabinovich et al., 2000]. The equations of the network are given by

$$\begin{aligned} \dot{v_i} &= v_i - v_i^3/3 - w_i - s_i\,(v_i - \nu_{\min}) + I \\ \dot{w_i} &= \varepsilon(v_i - bw_i + a) \\ \dot{s_i} &= \varepsilon\big(\sum_j g_{ji} H\,(v_j) - s_i\big) \end{aligned} \tag{2}$$

Although the original purpose of this small network model, as presented in the aforementioned reference, is to illustrate heteroclinic dynamics between three coexisting saddle-type limit cycle, it is not difficult to show that each FHN component of this network possesses a funnel structure, hence this network is a simple example of three-node funnel network; the equations of every node of system (2) above are indeed very similar to those of system (1). This shows that funnel dynamics can be compatible with heteroclinic dynamics, with

the substantial difference that it does not relie upon asymptotic dynamics and the present of saddle-type attractors, but only on transient dynamics, therefore we conjecture that it will be more robust and amenable to analog implementation.